

Embedding Model Selection for Domain-Specific Retrieval-Augmented Generation: A Comparative Study on Indian Cultural Heritage Corpora

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Abstract

Purpose: The performance of a Retrieval-Augmented Generation system is heavily dependent on the sentence embedding architecture used for dense retrieval. No previous research work has explored the effectiveness of various sentence transformer architectures on Indian heritage corpora and studied the effects of scaling corpus size on retrieval effectiveness in this context. This study attempts to fill those gaps.

Methods: We propose Kashivani, which is a modular RAG framework for cultural knowledge retrieval of Varanasi, India. An annotated corpus consisting of 15 documents containing 2,876 chunks is generated. Three different sentence transformer architectures – **all-MiniLM-L6-v2**, **all-mpnet-base-v2**, and **paraphrase-multilingual-MiniLM-L12-v2** – are tested with corpus scales of 638 and 2,876 chunks through an experiment with 36 question-answer pairs. Source Accuracy and Keyword Accuracy have been considered as retrieval quality metrics.

Results: The monolingual models consistently perform better than the multilingual model in terms of Keyword Accuracy by up to 20 percentage points. Corpus scaling changes the relative ranking of embedding models, with **all-mpnet-base-v2** performing best on the smaller corpus while **all-MiniLM-L6-v2** generalises better under larger-scale retrieval.

Conclusion: Monolingual sentence transformers outperform multilingual alternatives for English-language Indian cultural heritage retrieval. Model size is not a reliable proxy for retrieval quality under corpus scaling. Evaluation at the target corpus scale is necessary for informed embedding model selection in domain-specific RAG deployments.

Keywords: Retrieval-Augmented Generation, Sentence Transformers, Information Retrieval, Cultural Heritage, Indian NLP, Embedding Model Evaluation

1 Introduction

Retrieval-Augmented Generation (RAG) has become a key foundation for tying LLM-generated responses to authoritative knowledge bases [1]. The separation of document retrieval from the generation process leads to improvements in factual consistency and reduces hallucinations compared to purely parametric language models [2]. At the core of all RAG models lies the ability to retrieve the correct context, which is contingent on the quality of the underlying sentence embedding model that can encode both documents and queries into the same vector representation.

Models based on the Sentence-BERT framework [3] have become ubiquitous in the context of dense retrieval for RAG systems. Sentence transformers differ widely in terms of their architectural design, size of the model, and learning task. Some models are optimised for cross-lingual similarity retrieval using multilingual data (e.g., **paraphrase-multilingual-MiniLM-L12-v2**), while others focus on English-only representations (e.g., **all-MiniLM-L6-v2**, **all-mpnet-base-v2**). While model selection remains crucial for successful RAG deployments, there is a dearth of literature on a comparative study of sentence transformers for domain-specific contexts.

The retrieval of cultural heritage text is another unique information retrieval task. Cultural texts include the usage of historical language, religious language, Sanskrit-based words, and specialized named entities. In particular, Indian cultural heritage collections do not attract enough research interest within the RAG evaluation studies, despite their rich and varied content and the possibilities to use these intelligent retrieval systems not only for research but also in the general case.

To fill this gap, we develop and test Kashivani, which is the RAG system dedicated to retrieving cultural knowledge about Varanasi, India. Varanasi is one of the oldest continuously inhabited cities in the world [4] and covers such a wide range of topics as religion, classical music, textile culture, Buddhist pilgrimage history, and linguistics of the area. Our collection includes 15 documents containing over 286,000 characters in total and consisting of 2,876 text fragments. The performance of three types of sentence transformers is tested using 36 question-answer pairs.

Our contributions are as follows:

- We present Kashivani, a complete RAG pipeline built at the component level including document ingestion, overlapping chunking, FAISS-based vector indexing, semantic retrieval, and grounded LLM generation using LLaMA 3.1.
- We introduce a domain-specific evaluation framework for Indian cultural heritage retrieval comprising 36 manually validated question-answer pairs with ground-truth source attribution and keyword annotations.
- We conduct a systematic comparison of three sentence transformer architectures — **all-MiniLM-L6-v2**, **all-mpnet-base-v2**, and **paraphrase-multilingual-MiniLM-L12-v2** — across corpus scales of 638 and 2,876 chunks, measuring source retrieval accuracy and keyword precision.
- We report the empirical finding that monolingual models consistently outperform the multilingual model across both scales, that the performance gap widens with corpus size, and that model ranking reverses under scaling — with **all-MiniLM-L6-v2** overtaking **all-mpnet-base-v2** as the corpus grows.

The remainder of this paper is organised as follows. Section 2 reviews related work on RAG systems, sentence transformer evaluation, and cultural heritage information retrieval. Section 3 describes the system architecture of Kashivani. Section 4 presents the evaluation methodology. Section 5 reports experimental results. Section 6 discusses findings and their implications. Section 7 concludes with directions for future work.

2 Related Work

2.1 Retrieval-Augmented Generation

The Retrieval-Augmented Generation framework was proposed by Lewis et al. [1]. They showed that integrating a pre-trained seq2seq model and a dense vector indexing of Wikipedia into a unified system achieved superior performance on knowledge-intensive NLP tasks in comparison to fully parametric models. The basic premise that memory, not based on learning parameters, when queried via dense retrieval, alleviates hallucinations and enables better factual consistency was

proven in multiple applications and architecture variants [2, 5].

Several improvements have since been developed for each stage of the RAG pipeline separately. The importance of choosing an embedding model in the retrieval process was highlighted by a recent comparative analysis [6]. Other design choices like chunking strategy, overlap size, and top-k selection were explored as well and shown to be important for building an effective system alongside high-quality embeddings [7]. A new evaluation framework, RAGAS, offered a standardized way to evaluate RAG systems based on reference-free metrics such as faithfulness and answer relevance [8].

Nevertheless, a comprehensive assessment of RAGs using corpora specific to cultural domains that are underrepresented in terms of literature is still lacking. The majority of benchmark datasets utilized in RAG studies, such as those from the BEIR benchmark [9], come from English knowledge domains that belong to Western culture. As a result, there is an evident disparity in assessing RAGs when they are applied to the Indian cultural domain due to the difference in vocabulary and entities.

2.2 Sentence Transformer Architectures

The Sentence-BERT (SBERT) architecture [3] by Reimers and Gurevych set up the basis for contemporary dense retrieval methods by harnessing siamese and triplet networks for fine-tuning BERT models on semantic similarity tasks. The `all-MiniLM-L6-v2` model is distilled from a more extensive teacher model and creates a compact 22M parameter model that uses 384-dimensional embedding, optimized for efficiency and general purpose retrieval of English text. The `all-mpnet-base-v2` model originates from Microsoft’s MPNet architecture [10] and utilizes 768 dimensional embeddings, providing better results in semantic textual similarity benchmarks but at the cost of increased compute costs.

Multilingual sentence transformer models, like `paraphrase-multilingual-MiniLM-L12-v2`, extend the SBERT method into over 50 languages through multilingual training on parallel data sets. Litschko et al. [11] showed through a systematic cross-lingual retrieval experiment that although

the multilingual models have been successful for cross-lingual tasks, they often fail to achieve better results than specialized monolingual models in single language retrieval.

Although the Massive Text Embedding Benchmark (MTEB) [12] provides benchmark evaluation for 58 diverse tasks and many different languages, it is well known that the effectiveness of the model relies on the domain and type of the task. However, Indian culture datasets have no place in MTEB benchmark.

2.3 RAG for Cultural Heritage and Domain-Specific Knowledge

Limited but increasing attention has been paid to the use of RAG in cultural heritage domains. In a study by Liang et al. [13], an evaluation of RAG models for Hakka cultural knowledge retrieval was done using the method of cognitive benchmarking. It shows the superiority of the retrieval architecture on factual questions over the generation architecture. It proves the usefulness of knowledge grounding in cultural AI applications but does not take into account model selection and corpus size consideration.

Additionally, RAG has found uses in art history through provenance analysis [14]. The study proves the advantage of using RAG compared to general LLMs in such tasks, showing the usefulness of retrieval integration in such domains. However, such studies indicate the need for adaptation of evaluation methods to domain characteristics and appropriate embedding model selection.

As far as we know, no study has been done concerning the use of sentence transformers in Indian cultural heritage retrieval or the influence of corpus size. This paper addresses both gaps.

2.4 Vector Indexing for Dense Retrieval

The efficient computation of the similarity between vectors in a retrieval-augmented generation framework often uses an approximate nearest neighbor library. The Facebook AI Similarity Search (FAISS) library [15] offers both exact and approximate nearest neighbor searches on dense vector indexes that also include the possibility to compute L2 and inner product distances between

vectors. We chose to use the `IndexFlatL2` indexing method that performs an exact exhaustive search and is suitable for corpora containing thousands of vectors where accuracy is valued above speed.

3 System Architecture

Kashivani is a modular RAG architecture that is composed of implementation at component level without depending on any higher-level framework for the orchestration of RAG tasks. Such a design was intentional since each architectural decision is made at an independent component level, making them reproducible and experimentally testable. There are five consecutive stages in the Kashivani pipeline, as shown in Fig. 1, namely: document acquisition, text segmentation, vector indexing, semantic search, and grounding.

3.1 Document Ingestion

The ingestion module uses the `pypdf` package to extract text from PDF documents stored in a predefined corpus directory. Each document is then processed on a page-by-page basis, extracting text and concatenating it in one string. Unicode normalisation removes non-ASCII characters generated in the PDF conversion process, while regular expressions are used to replace sequences of spaces in the text. The result of this processing phase is a list of document objects, which consists of filename and cleansed text.

The resulting corpus contains 15 documents, spanning 286,000 characters across the following topics: urban and cultural history of Varanasi, textile heritage (Banarasi sari), Buddhist pilgrimage destinations (Sarnath), temple architecture (Kashi Vishwanath), Hindustani music (Benares gharana), regional linguistics (Bhojpuri, Hindi, Sanskrit), institutional knowledge (Banaras Hindu University), fort architecture (Ramnagar Fort), devotion literature (Tulsi Manas Mandir), regional politics (Uttar Pradesh), and religious philosophy (Hinduism). This corpus topic variety is deliberate in order to include heterogeneous vocabulary and named entities drawn from Sanskrit.

3.2 Chunking Strategy

Long documents cannot be fed into language models directly since it exceeds the limit of the context

window and dense retrieval requires high precision. The implementation of chunking uses the sliding window algorithm where each long document is divided into chunks by using character-level segmentation. Each chunk is divided into blocks consisting of 500 characters with an overlapping of 50 characters between each chunk. This helps preserve sentences that may otherwise be split across adjacent chunks.

The chunks are stored as objects consisting of a unique chunk id, file name from where the chunk was taken, text of the chunk, and a pair of character offsets in the source document. This helps identify the correct source of the chunk in our evaluation process as source accuracy is considered in evaluating the results. Chunking of 15 documents in the corpus yielded 2,876 chunks having an average text length of 487 characters per chunk.

3.3 Vector Indexing

Every text fragment is encoded into dense vectors via the use of sentence transformers architecture. In this study, we test three such architectures which have been outlined in Table 1. The embeddings are calculated in batches of 32 via `sentence-transformers` [3], with each batch being saved in 32-bit floating points.

FAISS `IndexFlatL2` [15] is used to index the resulting embedding matrix, and uses exact exhaustive nearest-neighbor search on the basis of squared Euclidean distance. An approximate approach like `IndexIVFFlat` was avoided since our dataset size of 2,876 vectors can be indexed with exact methods without making any computation trade-off, and since exact indexing will eliminate any indexing errors from affecting the quality of model retrieval. The index and its metadata file, containing the mapping of vector indexes to their chunk texts and filenames, are persisted for future use.

3.4 Semantic Retrieval

At inference time, the user query in natural language is transformed into an embedding using the exact same sentence transformer that was used for constructing the index, thereby making sure that the query and the document embeddings share the same semantic space. Then, the embedding of the query is sent to the FAISS index, which yields the

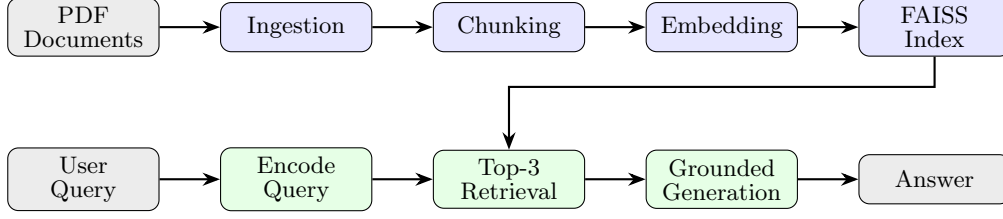


Fig. 1 Architecture of the Kashivani RAG pipeline. Top row (blue) shows the offline indexing phase. Bottom row (green) shows the online query phase. The FAISS index feeds retrieved chunks into the generation stage at query time.

Table 1 Sentence transformer models evaluated in this study

Model	Parameters	Dimensions	Type
all-MiniLM-L6-v2	22M	384	Monolingual
all-mpnet-base-v2	110M	768	Monolingual
paraphrase-multilingual-MiniLM-L12-v2	118M	384	Multilingual

indices and the L2 distance between the query and the k nearest chunk vectors. We use $k = 3$ in all of our experiments and retrieve the three semantically closest chunks. These chunks, along with their metadata, are then fed into the generation phase.

We set $k = 3$ because it ensures that we retrieve enough chunks without going over the context window limitations set by the generation model. Early experiments showed that $k = 3$ provided sufficient context for answering questions in a factual manner while not compromising on generation quality by retrieving extraneous information.

3.5 Grounded Generation

The generation module takes care of formatting the retrieved chunks and the user’s query to create an input prompt for the language model that is designed to only generate the response from the inputted context. The input prompt template is designed to ensure that the language model introduces itself as Kashivani, the guide to Varanasi, and only generates the response by strictly drawing its inspiration from the inputted context. If it does not get sufficient context from the inputs, it will reject generating any response.

The answers are created using the LLaMA 3.1 8B Instruct model available via Groq inference API. The temperature value is set at 0.2 to enable more accurate deterministic generation of the output. It has a maximum token limit of 500 tokens.

The rationale behind the choice of a low temperature value is determined by the fact that answering cultural heritage questions requires precision.

Table 2 summarises the key implementation parameters used across all experiments.

4 Evaluation Methodology

4.1 Research Questions

This study addresses three research questions:

- **RQ1:** Do monolingual sentence transformer models outperform multilingual models on source retrieval accuracy in an Indian cultural heritage corpus?
- **RQ2:** Does keyword precision differ significantly across sentence transformer architectures on the same corpus?
- **RQ3:** How does retrieval performance change as corpus size scales from a small corpus (638 chunks) to a larger corpus (2,876 chunks), and does the relative ranking of models change under scaling?

4.2 Corpus Construction

Evaluation was conducted across two corpus configurations to examine scaling behaviour:

- **Corpus A (Small):** 5 documents, 638 chunks, covering Varanasi city history, Banarasi sari, Ganga Aarti, Sarnath, and Kashi Vishwanath Temple.

Table 2 Key implementation and experimental configuration parameters

Parameter	Value
Chunk size	500 characters
Chunk overlap	50 characters
Top- k retrieval	3
Embedding batch size	32
Embedding models	Sentence Transformers
Vector index	FAISS <code>IndexFlatL2</code>
Generation model	LLaMA 3.1 8B Instruct
Temperature	0.2
Maximum generation length	500 tokens
Corpus sizes	638 and 2,876 chunks

- **Corpus B (Large):** 15 documents, 2,876 chunks, extending Corpus A with Banaras Hindu University, Benares gharana, Bhojpuri language, Hindi, Hinduism, Hindustani classical music, Ramnagar Fort, Sanskrit, Tulsi Manas Mandir, and Uttar Pradesh.

The augmentation from Corpus A to Corpus B was done to achieve linguistic and topic heterogeneity. These documents contain classical terms derived from Sanskrit, linguistic terms from the Bhojpuri language, music theory terms, and philosophical content. These augmentations provide linguistic and topic heterogeneity, which results in increased retrieval competition, meaning there is a chance of retrieving a chunk of text that might be semantically related but irrelevant to the actual document.

4.3 Evaluation Dataset

We constructed a manually validated evaluation dataset of 36 question-answer pairs. Questions were written to cover all 15 source documents proportionally, with each question targeting a specific factual claim present in exactly one source document. Each question is annotated with three fields:

- **expected_source:** the filename of the document containing the correct answer
- **expected_keywords:** a list of 2–4 domain-specific terms that must appear in retrieved chunks for the answer to be considered factually grounded

- **reference:** a ground-truth answer generated from the relevant source chunk using a zero-temperature LLaMA 3.1 8B model constrained to the source text

The data were further filtered such that the reference generator provided the response “no information in context”, implying that the chunk containing the keyword was not informative enough to generate a response. Such questions were filtered out from the dataset of 51 questions resulting in a final data set of 36 questions. Some examples are given in Table 3.

4.4 Evaluation Metrics

We define two retrieval quality metrics:

Source Accuracy (SA): The proportion of questions for which the expected source document appears in the top-3 retrieved chunks. Source accuracy measures whether the retrieval component identifies the correct document, independent of which specific passage within that document is returned.

$$SA = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\text{expected_source}_i \in \text{retrieved_sources}_i] \quad (1)$$

Keyword Accuracy (KA): The proportion of questions for which at least 50% of the expected keywords appear in the concatenated text of the top-3 retrieved chunks. Keyword accuracy measures semantic precision — whether the retrieved chunks contain the specific terminology required to answer the question correctly.

Table 3 Representative examples from the 36-question evaluation dataset

Question	Expected Source	Keywords
Name any two varieties of Banarasi sari?	Banarasi sari - Wikipedia.pdf	Katan, Shattir
Where did Buddha give his first sermon?	Sarnath - Wikipedia.pdf	Sermon, Buddha
What is moksha in Hindu philosophy?	Hinduism - Wikipedia.pdf	soul, liberation
What is the Benares gharana known for in music?	Benares gharana - Wikipedia.pdf	tabla, gharana
When was Banaras Hindu University established?	BHU - Wikipedia.pdf	1916

$$KA = \frac{1}{N} \sum_{i=1}^N \mathbb{1} \left[\frac{|\text{keywords}_i \cap \text{retrieved_text}_i|}{|\text{keywords}_i|} \geq 0.5 \right] \quad (2)$$

where $N = 36$ is the total number of evaluation questions, $\text{retrieved_sources}_i$ is the set of source filenames from the top-3 retrieved chunks for question i , and retrieved_text_i is the concatenated text of those chunks.

Both measures are complementary to each other. While source accuracy assesses the correctness of information retrieval at a very broad level (whether the relevant document was retrieved), keyword accuracy assesses the precision of information retrieval at a more detailed level (whether the relevant passage from the document was retrieved). A model can perform well at source accuracy but poorly at keyword accuracy.

Traditional information retrieval metrics such as Recall@k and Mean Reciprocal Rank (MRR) were not used because the evaluation focuses on domain-grounded factual retrieval rather than ranked document ordering. Since the objective is to determine whether retrieved chunks contain sufficient factual evidence for grounded answer generation, source-level and keyword-level relevance were considered more appropriate evaluation criteria.

4.5 Experimental Procedure

For each of the three sentence transformer models and each of the two corpus configurations, we:

1. Encode all chunks using the model to produce an embedding matrix
2. Build a FAISS IndexFlatL2 index over the matrix
3. For each of the 36 evaluation questions, encode the query using the same model and retrieve the top-3 nearest chunks

4. Compute source accuracy and keyword accuracy over all 36 questions

Six experimental conditions are created by this method: 3 models x 2 corpus sizes. The FAISS index is separately constructed for each experiment to make sure that there are no artifacts from the index construction process when measuring embeddings’ performance. All experiments were conducted on a CPU-based system (Intel i5, 16 GB RAM) to ensure hardware accessibility and reproducibility.

5 Results

5.1 Retrieval Performance on Corpus A (Small Corpus)

Table 4 presents source accuracy and keyword accuracy for all three sentence transformer models evaluated on Corpus A, comprising 638 chunks from 5 documents.

As regards Corpus A, while the two monolingual models manage to have an exact score in source accuracy of 100%, i.e., they always correctly find the source documents for each of the 36 questions posed through the top-3 retrieved chunks; the multilingual model scores lower source accuracy of 95.0%, as it fails to return the source document in only 2 cases out of the total 36 questions asked.

As far as keyword accuracy is concerned, it is clear that there is a greater difference between the three different models used for searching keywords in the corpus. While **all-mpnet-base-v2** is the best with respect to keyword accuracy with a score of 90.0% (it finds keyword containing passages in 32 out of 36 cases), the second position goes to **all-MiniLM-L6-v2** with a score of 75.0%. On the other hand, the multilingual model obtains the lowest score of 70.0%.

Table 4 Retrieval performance on Corpus A (5 documents, 638 chunks, 36 evaluation questions)

Model	Source Accuracy (%)	Keyword Accuracy (%)
all-MiniLM-L6-v2	100.0	75.0
all-mpnet-base-v2	100.0	90.0
paraphrase-multilingual-MiniLM-L12-v2	95.0	70.0

5.2 Retrieval Performance on Corpus B (Large Corpus)

Table 5 presents results for all three models evaluated on Corpus B, comprising 2,876 chunks from 15 documents.

Under corpus scaling, all three models experience degradation in source accuracy relative to Corpus A results. **all-MiniLM-L6-v2** declines from 100.0% to 96.1%, **all-mpnet-base-v2** declines from 100.0% to 94.1%, and the multilingual model declines from 95.0% to 88.2%. This universal degradation is consistent with the increased retrieval competition introduced by the larger corpus — as the number of chunks grows from 638 to 2,876, the probability of a semantically similar but topically irrelevant chunk displacing the correct source in the top-3 results increases.

Keyword accuracy on Corpus B shows a different pattern. **all-MiniLM-L6-v2** improves from 75.0% to 82.4% under scaling, while **all-mpnet-base-v2** declines from 90.0% to 82.4%, converging to the same score. The multilingual model remains at 70.6%, proving to be the lowest-performing model on both metrics.

5.3 Cross-Scale Comparison

Table 6 presents a consolidated view of all results across both corpus configurations, enabling direct comparison of scaling effects per model.

Three patterns emerge from this consolidated view:

Pattern 1 – Monolingual superiority is uniform. Monolingual models outperform the multilingual model regardless of corpus scales and metrics. For Corpus A, the difference between the best-performing monolingual model **all-mpnet-base-v2** in terms of keyword accuracy (90.0%) and the multilingual model (70.0%) is 20 percentage points. In turn, for Corpus B, the difference is reduced but remains the same: the best-performing monolingual model scores 82.4%

in keyword accuracy versus 70.6% for the multilingual model, which is 11.8 points apart.

Pattern 2 – Models switch positions when scaled. For Corpus A, **all-mpnet-base-v2** takes first place in keyword accuracy (90.0% vs 75.0% for **all-MiniLM-L6-v2**). On Corpus B, the results are tied with both models showing identical keyword accuracy (82.4%); however, **all-MiniLM-L6-v2** wins the second metric by gaining an additional 2.0 percentage points of source accuracy (96.1% vs 94.1%). This means that the better-performing model loses its edge as the retrieval competition intensifies, while the smaller model generalises better to the increased corpus size.

Pattern 3 – Multilingual degradation is disproportionate. The multilingual model experiences the largest absolute decline in source accuracy under scaling (6.8 percentage points, from 95.0% to 88.2%), compared to 3.9 points for **all-MiniLM-L6-v2** and 5.9 points for **all-mpnet-base-v2**. This suggests that the multilingual model is more sensitive to corpus growth, possibly because its multilingual training introduces greater semantic ambiguity in the embedding space when applied to English-only cultural heritage text.

6 Discussion

6.1 Monolingual Models for Domain-Specific English Retrieval

The superior performance of monolingual models compared to the multilingual model consistently across corpus sizes and metrics is in line with previous observations in the cross-lingual information retrieval literature. As shown in [11], multilingual encoders tend to fall short of specialized monolingual models on single-language information retrieval, with the trade-off in capacity due to

Table 5 Retrieval performance on Corpus B (15 documents, 2,876 chunks, 36 evaluation questions)

Model	Source Accuracy (%)	Keyword Accuracy (%)
all-MiniLM-L6-v2	96.1	82.4
all-mpnet-base-v2	94.1	82.4
paraphrase-multilingual-MiniLM-L12-v2	88.2	70.6

Table 6 Cross-scale comparison of retrieval performance across corpus configurations. SA = Source Accuracy, KA = Keyword Accuracy.

Model	Corpus A (638 chunks)		Corpus B (2,876 chunks)	
	SA (%)	KA (%)	SA (%)	KA (%)
all-MiniLM-L6-v2	100.0	75.0	96.1	82.4
all-mpnet-base-v2	100.0	90.0	94.1	82.4
paraphrase-multilingual-MiniLM-L12-v2	95.0	70.0	88.2	70.6

the use of one set of parameters for multiple languages identified as the cause. The contribution of this paper is the extension of such a result to a new domain of inquiry, namely Indian cultural heritage texts in English.

The difference in keyword accuracy on Corpus A of 20 percentage points between **all-mpnet-base-v2** and the multilingual model is much greater than expected, considering that both models have similar architectures and differ only in terms of their training datasets. The assumption is made that the domain-specific vocabulary of the corpus containing Sanskrit words and transliterations like *Dashashwamedh*, *Dhamek*, and *zari* is processed more accurately by English-only trained models. Multilingual training may compress the embedding space in ways that reduce the discriminative distance between semantically distinct cultural heritage terms that co-occur across languages in training corpora.

These findings have practical implications for system designers deploying RAG pipelines on Indian cultural heritage content. Where queries and documents are both in English, a monolingual model should be preferred over a multilingual alternative even when the domain involves Indian cultural concepts. The multilingual model’s advantage would likely materialise only in cross-lingual settings — for example, Hindi-language queries against English documents — which represents a natural direction for future work.

6.2 Corpus Scaling and Model Ranking Reversal

The flipping of ranking order when scaling corpora is the most counterintuitive discovery of this experiment. While **all-mpnet-base-v2** dominates **all-MiniLM-L6-v2** in terms of keyword accuracy on Corpus A by 15 percentage points, on Corpus B, both models exhibit similar keyword accuracy scores while the smaller model beats out the larger one in source accuracy. This finding opposes the conventional understanding that increased embedding size equates to enhanced robustness in retrieval when faced with increased corpus complexity.

One possible reason behind this phenomenon has to do with the type of corpus growth employed in this experiment. The addition of 10 documents brings linguistic diversity into the fold, such as Sanskrit-related terms, Bhojpuri-related phrases, music theory terminology, etc., making for a higher occurrence of semantically related yet topically dissimilar chunks of data. Under these circumstances, a 768-dimensional embedding produced by **all-mpnet-base-v2** might have mapped topically similar but culturally different concepts into proximity to each other in a high-dimensional embedding space, causing confusion during retrieval processes.

The above interpretation is speculative and needs to be verified through the use of space visualization techniques, such as embedding space visualization using t-SNE where embeddings for

different chunks are colored based on their source documents. This will be covered as an avenue for further research. The findings from this study, however, are that model size and dimensionality of the embedding do not serve as good metrics to measure retrieval performance in scaling conditions within a particular domain.

6.3 Universal Retrieval Degradation Under Scaling

The consistent drop in source accuracy for all three models as the corpus grows from 638 chunks to 2,876 chunks is also in line with theoretical predictions regarding dense retrieval systems. The larger the corpus, the larger the number of chunks vying for the top k spots, and therefore, the higher the likelihood that some semantically similar, but thematically unrelated, chunk would displace the correct source. This phenomenon is known to be an issue with flat exhaustive search indices such as `IndexFlatL2`, prompting further research into advanced retrieval techniques like re-ranking, sparse-dense hybrid retrieval, and domain-specific embedding tuning for increasingly large corpora.

The excessive degradation of the multilingual model (6.8 percentage points lower accuracy compared to 3.9 and 5.9 points for monolingual models) could hint at the vulnerability of multilingual embeddings when it comes to retrieval errors in culturally heterogeneous collections. As the collection grows and becomes more diverse in its sub-domains and their common cultural vocabulary, the multilingual model’s compressed representations are prone to mixing up various concepts.

6.4 Limitations

It is worth noting that several aspects have been limiting factors in the current research. First, the size of the evaluation dataset consisting of 36 questions is considerably smaller compared to the standard IR benchmark datasets. No statistical significance test was done for any differences seen between the results achieved by different models, which might mean that certain differences seen may not be significant enough; for instance, the 2.0 percentage points source accuracy difference between the two models on Corpus B.

Second, the fact that the corpus is made up of exclusively English Wikipedia articles means that generalisability to sources such as archived documents, transcripts of oral history, and Hindi primary sources might not be possible. It remains to be seen whether the multilingual advantage shown for Hindi queries is inverted on queries made to the same corpus containing both languages as the current evaluation setting does not allow this.

Third, the faithfulness and answer relevancy metrics were not calculated owing to the lack of compatibility between RAGAS v0.4.3 evaluation codebase and open-source LLM inference providers.

Fourth, all three models were used in their pre-trained, non-fine-tuned form. Fine-tuning on a domain-specific Indian cultural heritage dataset would likely improve performance for all models and may alter the relative ranking observed in this study.

7 Conclusion

In this paper, we introduce Kashivani, a modular Retrieval Augmented Generation model for cultural knowledge retrieval regarding Varanasi, India. We conduct an analysis of three different sentence transformer architectures on two different scales of the corpus. Our experiment involving a manually validated set of 36 question-answer pairs from 15 Indian cultural heritage documents results in three key findings.

Our first finding is that monolingual sentence transformers outperform the multilingual model on both corpus scales as well as on both evaluation metrics. In particular, on a small corpus, there is a 20 percentage point difference in keyword accuracy between the best monolingual model and the multilingual model which indicates that semantic inconsistencies introduced by multilingual training hurt performance on the target language corpus. Our findings contribute to cross-lingual retrieval literature [11] by demonstrating this phenomenon for the new domain of Indian cultural heritage texts written in English.

Our second key finding is that model ranking is inverted between corpus sizes. While `all-mpnet-base-v2` outperforms on keyword accuracy in the small corpus setting,

all-MiniLM-L6-v2 equals or outperforms it while having higher source accuracy on a larger corpus.

Thirdly, all models show an inevitable source accuracy degradation as the corpus size increases from 638 to 2,876 chunks, with the multilingual model suffering the biggest absolute decline in terms of percentage points, at 6.8 percentage points. This demonstrates that evaluations should be done using the corpus size at deployment instead of a smaller dataset.

This analysis has implications for the deployment of RAGs in Indian cultural heritage by practitioners. Based on the results obtained, we recommend all-MiniLM-L6-v2 embedding model for English queries in Indian cultural heritage RAG systems with corpus sizes of more than 2,000 chunks due to its better scalability properties.

The next step of future research includes three parts. Firstly, extending the current study to evaluating Hindi queries to see whether the multilingual model recovers its superiority in cross-lingual retrieval due to the wide coverage of its training data. Secondly, performing domain-specific fine-tuning of the models used in this paper on Indian cultural heritage texts may enhance the absolute scores as well as influence their relative ranking. Thirdly, introducing generation quality metrics such as faithfulness and answer relevancy in the evaluation framework.

Declarations

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Competing interests: The authors have no competing interests to declare that are relevant to the content of this article.

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Consent to participate: Not applicable.

Consent for publication: All authors have read and approved the final manuscript and consent to publication.

Data availability: The evaluation dataset of 36 question-answer pairs and corpus document list are available from the corresponding author

upon reasonable request. The source documents are publicly available Wikipedia articles.

Code availability: The complete source code for the Kashivani pipeline including ingestion, chunking, embedding, retrieval, generation, and evaluation modules is available at: https://github.com/PriyankaAsthana/KashiVani_RAG

Author contributions: Priyanka Asthana conceptualised the study, designed and implemented the complete pipeline, constructed the evaluation dataset, conducted all experiments, and drafted the manuscript. Dr. Laxmi Shanker Maurya supervised the research, provided critical revision of the manuscript, and validated the methodology. Harshita Kushwaha contributed to corpus construction and evaluation dataset validation.

References

- [1] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.
- [2] Kurt Shuster, Spencer Poff, Moya Chen, Douwe Kiela, and Jason Weston. Retrieval augmentation reduces hallucination in conversation. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 3784–3803, 2021.
- [3] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 3982–3992, 2019.
- [4] Wikipedia contributors. Varanasi — wikipedia, the free encyclopedia, 2026. Accessed: 2026-05-25.
- [5] Yunfan Gao, Yun Xiong, Xinyu Gao, Kaijie Jia, Jinlei Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2023.
- [6] Zhangyin Chen et al. A comparison of independent and joint fine-tuning strategies

- for retrieval-augmented generation. *arXiv preprint arXiv:2510.01600*, 2024.
- [7] Antonio Jimeno Yepes, Yao You, Jan Milczek, Sebastian Laverde, and Renxi Li. Financial report chunking for effective retrieval augmented generation. *arXiv preprint arXiv:2402.05131*, 2024.
 - [8] Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. RAGAS: Automated evaluation of retrieval augmented generation. *arXiv preprint arXiv:2309.15217*, 2023.
 - [9] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2021.
 - [10] Kaitao Song, Xu Tan, Tao Qin, Jianfeng Lu, and Tie-Yan Liu. MPNet: Masked and permuted pre-training for language understanding. In *Advances in Neural Information Processing Systems*, volume 33, pages 16857–16867, 2020.
 - [11] Robert Litschko, Ivan Vulić, Simone Paolo Ponzetto, and Goran Glavaš. On cross-lingual retrieval with multilingual text encoders. *Computational Linguistics*, 48(3):607–651, 2022.
 - [12] Niklas Muennighoff, Nouamane Tazi, Loïc Magne, and Nils Reimers. MTEB: Massive text embedding benchmark. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2014–2037, 2023.
 - [13] Chen Liang et al. Evaluating cultural knowledge processing in large language models: A cognitive benchmarking framework integrating retrieval-augmented generation. *arXiv preprint arXiv:2511.01649*, 2024.
 - [14] Giles Caspari et al. Retrieval-augmented generation for natural language art historical research. *arXiv preprint arXiv:2508.19093*, 2024.
 - [15] Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547, 2019.